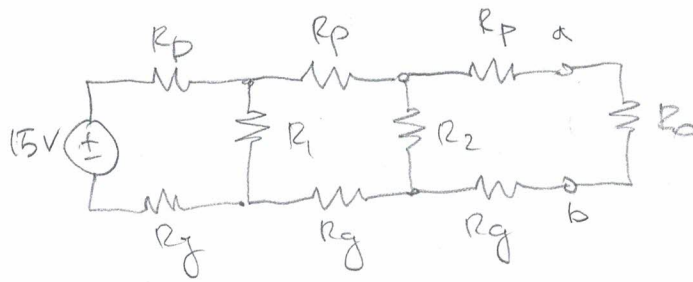


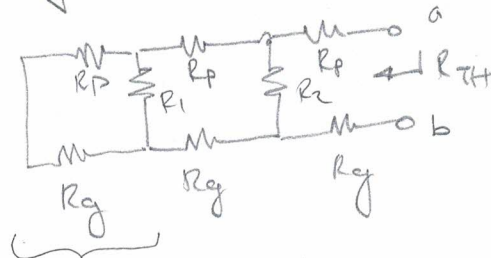
3.58)



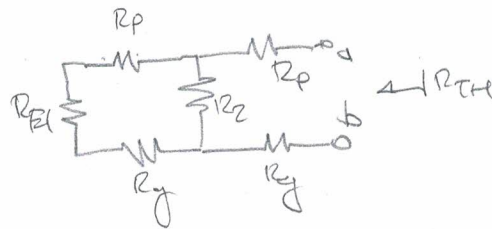
BESTIMMEN
THEVENINENKV. fÜR
POLENA A UND B.

$$\begin{cases} R_P = 1 \Omega \\ R_g = 0.1 \Omega \\ R_1 = 10 \Omega \\ R_2 = 20 \Omega \end{cases}$$

Lösung: KORTSLOT SP. NUTZEN



$$R_{TH1} = \frac{R_1(R_P + R_g)}{R_1 + R_P + R_g} = \frac{10(1.1)}{10 + 1.1} \approx 0.991 \Omega$$

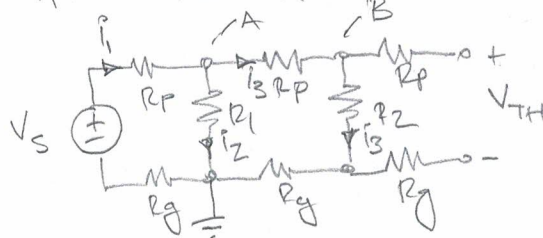


$$R_{TH2} = \frac{R_2(R_P + R_g + R_{TH1})}{R_2 + R_P + R_g + R_{TH1}} \approx 1.893 \Omega$$

SAMT

$$R_{TH} = R_P + R_g + R_{TH2} = 1 + 0.1 + 1.893 \approx 2.993 \Omega$$

ÄRTERSTÄR V_{TH} (KORTSCHWINGUNGSSPANNUNGEN). V_1 UÄR



NORMANALYS fÜR NOD A UND B.

$$A: i_1 - i_2 - i_3 = 0 \quad (1)$$

$$B: i_3 - i_3 = 0 \quad (2)$$

SAMT

$$-R_g i_1 + V_S - R_P i_1 = V_A \quad (3)$$

UTRICKEL STRÖM I TERMER AV
SPÄNNING

$$i_2 = \frac{V_A}{R_1}, \quad i_3 = \frac{V_B}{R_2 + R_g} \quad (4)$$

(3) ger

$$i_1 = \frac{-R_g i_1 + V_S - V_A}{R_P} \Rightarrow i_1 = \frac{V_S - V_A}{R_P + R_g}$$

(1) KCL

$$\frac{V_S - V_A}{R_P + R_g} - \frac{V_A}{R_1} - \frac{V_B}{R_2 + R_g} = 0$$

(2) KCL

$$\frac{V_B}{R_2 + R_g} - \frac{V_A - V_B}{R_P} = 0$$

matrix form

$$\begin{bmatrix} -\frac{1}{R_P + R_g} - \frac{1}{R_1} & -\frac{1}{R_2 + R_g} \\ -\frac{1}{R_P} & \frac{1}{R_2 + R_g} + \frac{1}{R_P} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} -\frac{V_S}{R_P + R_g} \\ 0 \end{bmatrix}$$

with numbers

$$\begin{bmatrix} -\frac{1}{1.1} - \frac{1}{10} & -\frac{1}{20.1} \\ -\frac{1}{1} & \frac{1}{20.1} + \frac{1}{1} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} -\frac{15}{1.1} \\ 0 \end{bmatrix}$$

MATLAB KCL

$$\begin{cases} V_A = 12.71 \text{ V} \\ V_B = 12.30 \text{ V} = V_{TH} \end{cases}$$

SVAR: THEVENIN/EKV/VALENTIN
KCL

